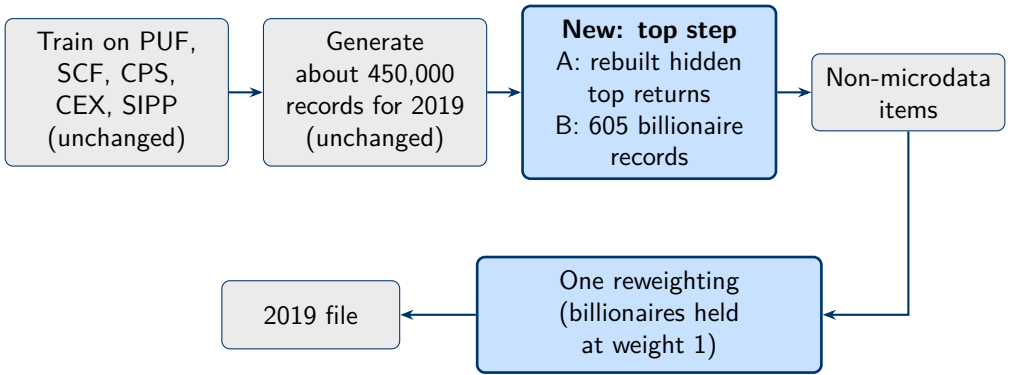
The PUF's aggregate records and the billionaires in the Tax-Data rebuild

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The rebuild in one picture



Two things change: about 2,800 records are added before the reweighting, and the billionaires' weights cannot move.

Problem 1: the PUF's aggregate records

In the 2015 PUF, SOI folded 1,233 extreme returns into four records that hold group means. Together they hold $140 billion of AGI.

AGI group	Returns	Total AGI
Negative AGI	140	−$13.5bn
$0–10 million	233	$0.9bn
$10–100 million	394	$17.2bn
Over $100 million	465	$135.7bn

The retained PUF cannot reach SOI's top 0.001% by AGI: its largest AGI, $136 million, is below that group's mean, $152 million.

2015 dollars.

Problem 2: the billionaires, and why not add them before training

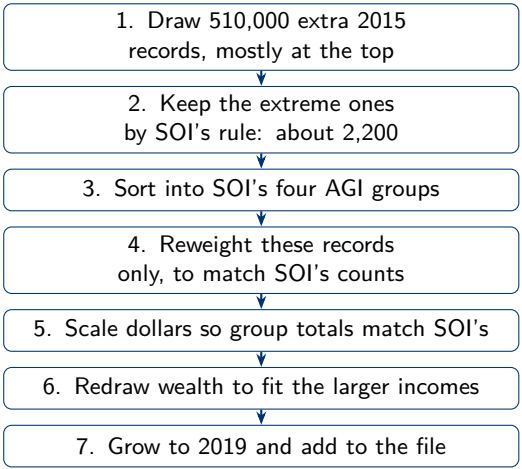
The SCF drops the Forbes 400 by name. The PUF hides them in its aggregate records.

Adding them to the training data would not work:

- A few hundred rows among about 136,000 SCF rows would barely move the network.
- The list gives wealth but no tax items to learn from.
- Drawn dollar amounts cannot exceed the training data's largest values.

So both problems are handled after generation, before the reweighting.

Part A: rebuilding the aggregate records



The network supplies which items go together. SOI's published totals supply the sizes.

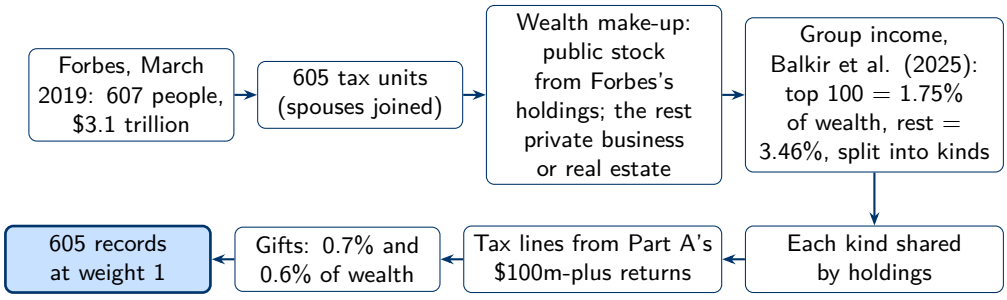
The method follows Ricco (2020) for the Penn Wharton Budget Model.

Part A: a worked example

Illustrative numbers.

1. **Weights.** SOI's over-$100-million record stands for 465 returns. We keep 600 records, so each starts at $465/600 = 0.775$. The weights are then nudged until SOI's counts match.
2. **Dollars.** SOI's group total of long-term gains is $80 billion. Our weighted records total $30 billion. Every record's gains in that group are multiplied by $80/30 = 2.67$.
3. Gains and losses are scaled separately. AGI is recomputed from the parts.

Part B: building the billionaire records

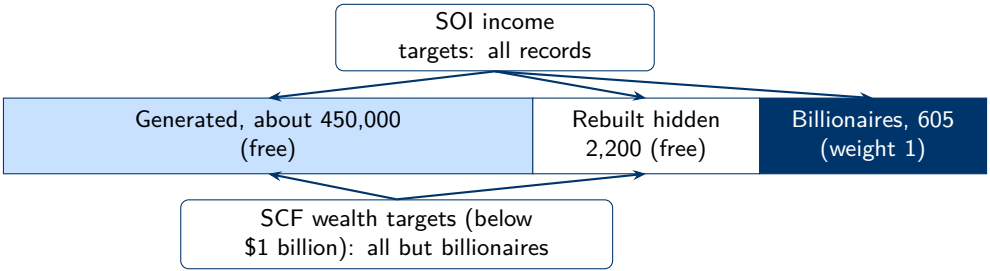


Part B: a worked example

Illustrative numbers.

1. The top group's wealth is \$1,000 billion. Its dividends total $1{,}000 \times 1.75\% \times 28\% =$ \$4.9 billion.
2. Dividends are shared in proportion to public stock.
3. Zuckerberg holds 15% of the group's public stock, so his dividends are $15\% \times$ \$4.9 billion $=$ \$0.74 billion.

Part C: one reweighting



- Example: the top 0.001% AGI target is $214.5 billion. The billionaires contribute a fixed $60 billion (illustrative), so the free records supply $154.5 billion.
- No double count: SOI's totals include the billionaires, whose share is subtracted; the SCF wealth targets stop at $1 billion.
- A check before solving stops the run if any target is out of reach.

Results

- 2015: top 0.001% AGI is $216 billion, against SOI's $215 billion. Without the rebuilt returns it is $103 billion.
- 2019: all 805 targets are met. Top 0.001% AGI is $212.3 billion, against SOI's $214.5 billion.
- Billionaire AGI per unit, top group: $329 million, against $345 million in the IRS study.
- Gifts per unit, top group: $133 million, against $140 million.
- The rebuilt over-$100-million returns fall from $151 billion to $65 billion of AGI to make room.
- Effective sample of the free records is unchanged at 80.3%.

Nominal dollars of 2015 and 2019 as labelled. Tested on the 28–29 October networks.

Limits

- A family fortune is kept as one record.
- Income ratios below Forbes rank 400 are extrapolated.
- The mix of items inside the hidden groups comes from the network. It is not observed.
- Replacing the SCF's households above $1 billion with the list is an assumption.
- The tax calculator has not yet been run on the billionaire records.